



PROJECT REPORT

URBAN AIR QUALITY INTELLIGENCE

Developed by :

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Developed as part of the **IBM SkillsBuild Data Analytics with AI Internship 2026.**

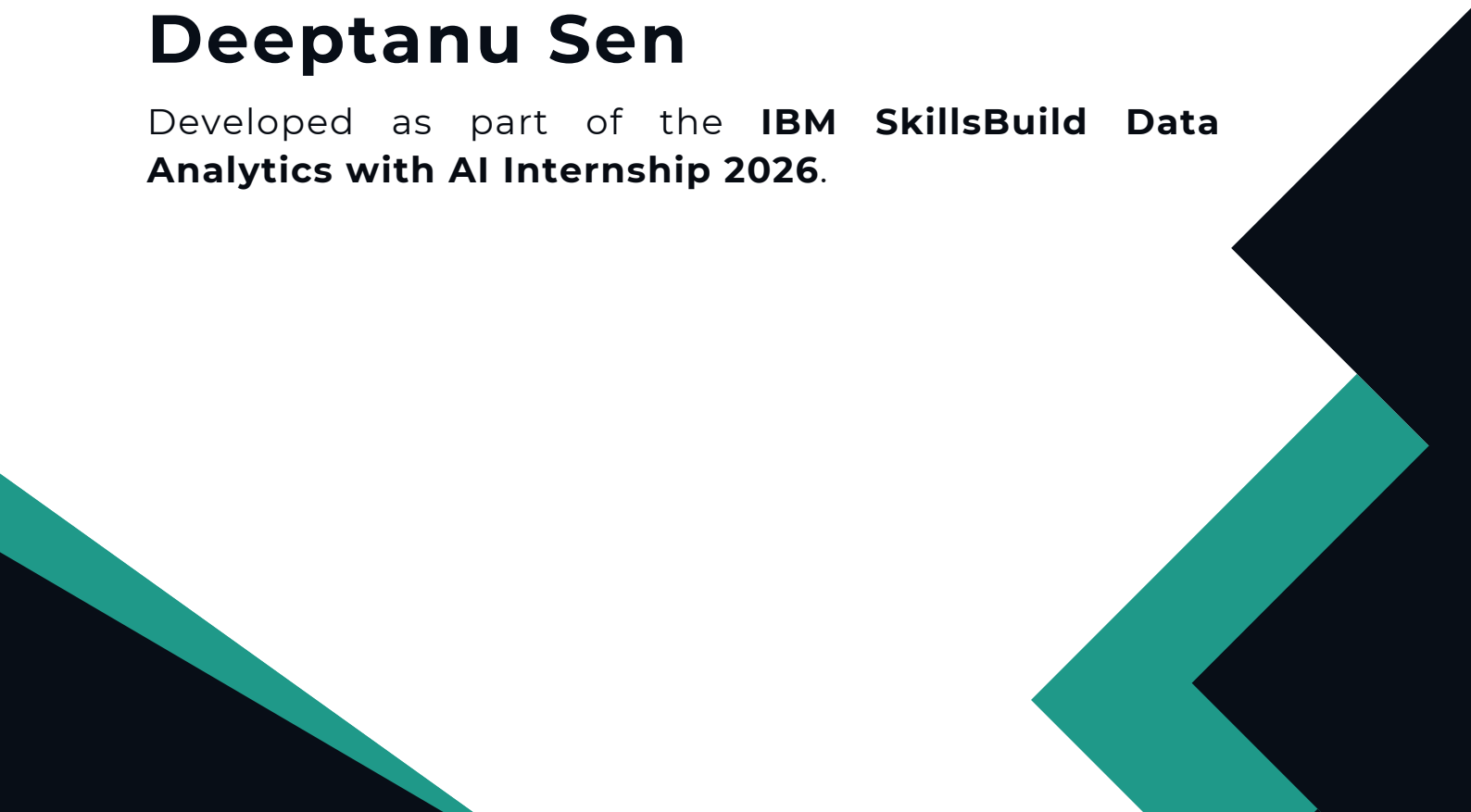


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Urban Air Quality Intelligence

(Descriptive and Diagnostic Analysis)

An interactive Streamlit dashboard for analysing urban air quality across 26 Indian cities using CPCB city-day measurements (2015–2020).

01 · Project Overview

Urban Air Quality Intelligence is an interactive Streamlit dashboard for exploring air-quality patterns across **26 Indian cities from 2015–2020**. Built using the CPCB city-day dataset, it combines descriptive statistics, interactive visualizations, and diagnostic analysis to examine **AQI trends, seasonal patterns, pollutant relationships, city-level differences, and high-risk periods** through a clean, filter-driven interface.

Results describe observed associations in the dataset. Correlation does not establish causation.

ANALYTICAL CAPABILITIES

Area	Methods & Features
AQI Analysis	Distribution · categories · yearly / monthly / seasonal trends
City Comparison	Rankings · city × year heatmaps
Pollutant Profiling	Distribution by city · normalised radar profiles · AQI relationships
Correlation	Pearson r · 13×13 matrix · per-pollutant p-values (AQI pairs)
Risk Analysis	$AQI \leq 100$ vs $AQI > 200$ · Mann–Whitney U test
Spike Detection	30-day rolling z-score · threshold > 2.0
Driver Analysis	Normalised pollutant contribution by city, year & season

02 · Dataset and Preparation

Dataset Overview

The dataset powering this project is **city_day.csv** - a publicly available air-quality measurements dataset compiled by the **Central Pollution Control Board (CPCB)** of India. It captures daily pollutant readings from official monitoring stations across **26 major Indian cities**, spanning over **five years from January 2015 to July 2020**.

AT-A-GLANCE SUMMARY

Attribute	Detail
Dataset Name	city_day.csv [Kaggle] or, [Google Drive]
Source	Central Pollution Control Board (CPCB), Government of India
Date Range	January 1, 2015 → July 1, 2020 (5.5 years)
Total Records	29,531 daily observations (rows)
Total Columns	16 (City, Date, 12 Pollutants, AQI, AQI_Bucket)

CITIES COVERED

North India	South India	East India	West India
Delhi	Chennai	Kolkata	Mumbai
Gurugram	Bengaluru	Patna	Ahmedabad
Lucknow	Hyderabad	Brajrajnagar	Jaipur
Amritsar	Kochi	Jorapokhar	Bhopal
Chandigarh	Thiruvananthapuram	Talcher	
	Coimbatore	Guwahati	
	Visakhapatnam · Amaravati	Shillong · Aizawl	

COLUMN SCHEMA

The dataset contains 16 columns categorised into identifiers, pollutant measurements, and AQI output fields. All pollutant concentrations are in $\mu\text{g}/\text{m}^3$ except CO, which is in mg/m^3

#	Column	Type
1	City	Categorical
2	Date	DateTime
3	PM2.5	Float ($\mu\text{g}/\text{m}^3$)
4	PM10	Float ($\mu\text{g}/\text{m}^3$)
5	NO	Float ($\mu\text{g}/\text{m}^3$)
6	NO2	Float ($\mu\text{g}/\text{m}^3$)
7	NOx	Float ($\mu\text{g}/\text{m}^3$)
8	NH3	Float ($\mu\text{g}/\text{m}^3$)
9	CO	Float (mg/m^3)
10	SO2	Float ($\mu\text{g}/\text{m}^3$)
11	O3	Float ($\mu\text{g}/\text{m}^3$)
12	Benzene	Float ($\mu\text{g}/\text{m}^3$)
13	Toluene	Float ($\mu\text{g}/\text{m}^3$)
14	Xylene	Float ($\mu\text{g}/\text{m}^3$)
15	AQI	Float
16	AQI_Bucket	Categorical

INDIA CPCB AQI SCALE

The Air Quality Index (AQI) used in this project follows the **official India CPCB scale**. This scale defines six bands, each associated with a health advisory. The dashboard uses these same bands for category-based filtering, colour coding, and diagnostic thresholds.

AQI Range	AQI Range	AQI Range	AQI Range
0 – 50	Good	Minimal impact	Green
51 – 100	Satisfactory	Minor breathing discomfort to sensitive persons	Light Green
101 – 200	Moderate	Breathing discomfort to asthma patients & elderly	Yellow
201 – 300	Poor	Breathing discomfort to most people on prolonged exposure	Orange
301 – 400	Very Poor	Respiratory illness on prolonged exposure	Red
401 – 500	Severe	Health impacts even on light physical activity; serious risk	Purple

Data Quality & Cleaning

The raw dataset required a structured cleaning pipeline before analysis. The pipeline addresses -

- Date parsing errors
- Exact duplicate records
- Extreme AQI instrument spikes
- Below-detection-limit zeros in VOC sensors
- Pervasive missing pollutant readings - which ranged from ~7% (CO) to over 60% (Xylene) in the original data.

CLEANING PIPELINE

Steps	Action	Records Affected
1. Parse Date	Converted to datetime; coerce errors → NaN, then dropped	0
2. Drop Duplicates	Removed exact (City, Date) duplicate rows	0
3. Cap Extreme AQI	AQI > 999 capped at 999 (instrument spike suppression)	93
4. Zero BTX/CO → NaN	Below-detection-limit zeros in Benzene, Toluene, Xylene, CO treated as missing	10,738
5. City-Month Imputation	Missing pollutants filled with city-month median (preserves seasonal & regional patterns)	All rows
6. City Median Fallback	Residual NaN filled with city-level median	All rows
7. Global Median Fallback	Cities with zero data for a pollutant filled with global dataset median	All rows
8. AQI Imputation	Missing AQI filled with city-month median AQI	All rows
9. AQI_Bucket Re-derived	Bucket recomputed from final imputed AQI values	All rows

KEY DATASET STATISTICS

After the full cleaning pipeline, the dataset retained all 29,531 records with zero remaining missing values in pollutant or AQI columns.

Metric	Value
Total Observations (clean)	29,531
Number of Cities	26
Date Range	2015-01-01 to 2020-07-01
Pollutants Tracked	12 (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene)
Overall Average AQI	165.2 (Moderate category)
Pollution Risk Rate (AQI > 200)	25.8% (7,605 of 29,531 days)
Dominant Pollution Driver	PM10 (highest normalised mean concentration)
Missing Values After Cleaning	0 across all pollutant and AQI columns

03 · Technology Stack

Python 3.11+

Streamlit

Plotly

Pandas

NumPy

SciPy

The project uses **Python** and **Streamlit** for application logic and the interactive dashboard, with **Pandas** and **NumPy** handling data processing and numerical analysis. **Plotly** provides interactive visualizations, while **SciPy** supports statistical tests and correlation analysis.

Technology	Role
Python	Application logic & analysis
Streamlit	Interactive dashboard framework
Pandas	Data loading, cleaning, aggregation
NumPy	Numerical operations & rolling statistics
Plotly	Interactive charts (bar, line, scatter, heatmap, radar, pie)
SciPy	Pearson correlation, Mann–Whitney U, p-values

04 · Dashboard

The **Urban Air Quality Intelligence dashboard** is a fully interactive, dark-mode web application built with **Streamlit**. It is designed around a single-page architecture: one persistent sidebar for global filtering controls and a main area that exposes **five distinct analytical modules**. Every chart, table, and KPI on the page is derived from the same filtered data slice - **adjusting any sidebar control instantly recomputes and re-renders the entire dashboard in real time**.

The layout follows a top-down narrative: the hero banner establishes context, **three headline KPIs deliver** the immediate story, a supporting stats strip adds depth, and the tab modules enable deep-dive exploration across **AQI, city, pollutant, diagnostic, and data quality dimensions**.



Figure 1: Urban Air Quality Intelligence - Live Dashboard Overview

HEADLINE KPI CARDS

Three KPI cards span the top of the main content area. Each card is driven entirely by the active filter selection and recomputes on every interaction:

KPI Card	Metric Definition	Dynamic Behaviour
Average AQI	Arithmetic mean of AQI across all filtered rows. Colour-coded badge shows the AQI category (Good → Severe).	Changes with every city / year / season selection
Pollution Risk Rate	% of days with AQI > 200 (India threshold for 'Poor' - first level with broad health impact). Shows raw count as a badge.	Risk level badge shifts colour: green < 10%, amber 10–25%, orange 25–50%, red ≥ 50%
Dominant Pollution Driver	Each pollutant's mean is normalised by its reference ceiling (NAAQS/WHO max). The pollutant with the highest normalised score is declared dominant - never hardcoded.	Can change city-to-city: PM10 for East India, Toluene for West India in winter

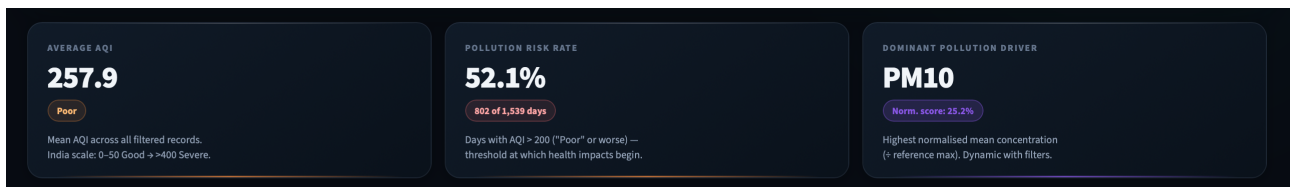


Figure 2: 3 KPI cards - showing a filtered view for 10 cities, 2015–2016, Winter

NAVIGATION TABS

The tab bar provides access to five analytical modules. All tabs operate on the same filtered dataset derived from the sidebar - no extra filtering is needed inside each tab.

Tab	What It Covers
AQI Intelligence	AQI distribution, category donut, value histogram, category breakdown table, worst pollution days, yearly/monthly trend lines, seasonal bar chart
City Intelligence	Per-city AQI ranking bar chart, city × year AQI heatmap, descriptive statistics table per city
Pollution Intelligence	Pollutant-AQI correlation bar, normalised pollution driver radar, seasonal pollutant boxplots, city-wise pollution distribution by pollutant
Diagnostic Intelligence	Auto-generated text insights, full correlation matrix, AQI spike & anomaly detection, normal vs. high-risk day comparison, pollutant-to-AQI drilldown, dominant driver shifts
Data Quality	Dataset overview stats, 9-step cleaning pipeline table, original missing value rate chart

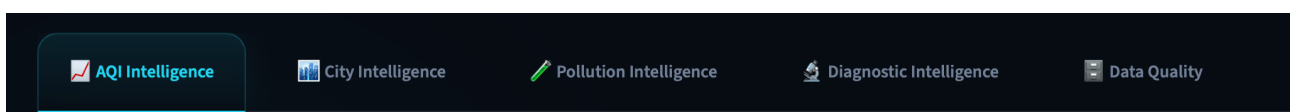


Figure 3: Navigation tab bar - 5 analytical tabs

• Analysis Controls - The Sidebar

The sidebar is the single source of truth for all filtering in the dashboard. Labelled **ANALYSIS CONTROLS** with the scope indicator 'INDIA · 2015 – 2020', it contains **four stacked filter controls** and a **live selection summary panel** at the bottom. Every control operates independently and can be combined - the dashboard always **reflects the intersection of all active filters** simultaneously.

FILTER CONTROLS

Control	Type	Behaviour & Options
City Selection	Preset dropdown + multi-select	Five regional presets: • All cities • North India • South India • East India • West India and Custom. Selecting a preset pre-fills the multi-select with the relevant cities. Individual cities can be added or removed freely.
Date Range	Dual-handle year slider	Slide both handles to restrict the analysis to any year window from 2015 to 2020. The slider label updates to reflect the selected range.
Seasons	Multi-select (4 options)	Winter (Dec–Feb), Spring (Mar–May), Summer (Jun–Aug), Autumn (Sep–Nov). Deselect any season to exclude it. Default = all four seasons selected.
AQI Category Focus	Multi-select (6 options)	Filter to only records matching specific AQI buckets (Good, Satisfactory, Moderate, Poor, Very Poor, Severe). Useful for studying a single pollution tier in isolation.

ANALYSIS CONTROLS

INDIA · 2015 – 2020

CITY SELECTION

All cities

Ahmedabad

Aizawl

Amaravati

Amritsar

Bengaluru

Bhopal

Brajrajnagar

Chandigarh

Chennai

Coimbatore

Delhi

Ernakulam

Gurugram

Guwahati

Hyderabad

Jaipur

Jorapokhar

Kochi

Kolkata

Lucknow

Mumbai

Patna

Shillong

DATE RANGE

2015

2020

SEASONS

Winter

Spring

Summer

Autumn

AQI CATEGORY FOCUS

Good

Satisfactory

Moderate

Poor

Very Poor

Severe

SELECTION SUMMARY

Records

29,531

Cities

26

Mean AQI

165

Period

Jan 2015 → Jul 2020

Figure 4: Full sidebar - all four filter controls and live Selection Summary panel

ANALYSIS CONTROLS

INDIA · 2015 – 2020

CITY SELECTION

All cities

All cities

North India (Delhi/Gurugram/Lucknow/Patna/Amritsar/Chandigarh)

South India (Chennai/Bengaluru/Hyderabad/Kochi/Thiruvananthapuram/Coimbatore)

East India (Kolkata/Patna/Brajrajnagar/Jorapokhar/Talcher/Guwahati/Shillong/Aizawl)

West India (Mumbai/Ahmedabad/Jaipur/Bhopal)

Custom

Figure 5: City Selection preset dropdown - regional zone presets

LIVE SELECTION SUMMARY PANEL

At the bottom of the sidebar, a Selection Summary panel provides immediate feedback on the active filter state. It displays four live metrics that update in real time as the user adjusts any filter:

Summary Metric	What It Shows
Records	Total number of rows in the filtered dataset
Cities	Count of unique cities in the current selection
Mean AQI	Average AQI across the filtered records
Period	Date span of the filtered data (e.g. Jan 2015 → Jul 2020)

DYNAMIC FILTER EXAMPLES

The sidebar's power becomes evident when filters are combined. The two examples below demonstrate how dramatically the KPIs shift when the same analysis is restricted to specific regions, years, and seasons - illustrating the dashboard's real-time contextual intelligence.

Example A - East India, Year 2017, Winter:

Cities: Kolkata, Patna, Brajrajnagar, Jorapokhar, Talcher, Guwahati, Shillong, Aizawl | Records: 158 | Avg AQI: 295.7 (Poor) | Risk Rate: 81.6% | Dominant Driver: PM10



Figure 6: Dynamic filter - East India preset, Year 2017, Winter season. AQI jumps to 295.7 with 81.6% risk rate and 26 Severe Days

Example B - West India, 2015–2016, Winter:

Cities: Mumbai, Ahmedabad, Jaipur, Bhopal | Records: 362 | Avg AQI: 301.5 (Very Poor) | Risk Rate: 49.4% | Dominant Driver: Toluene

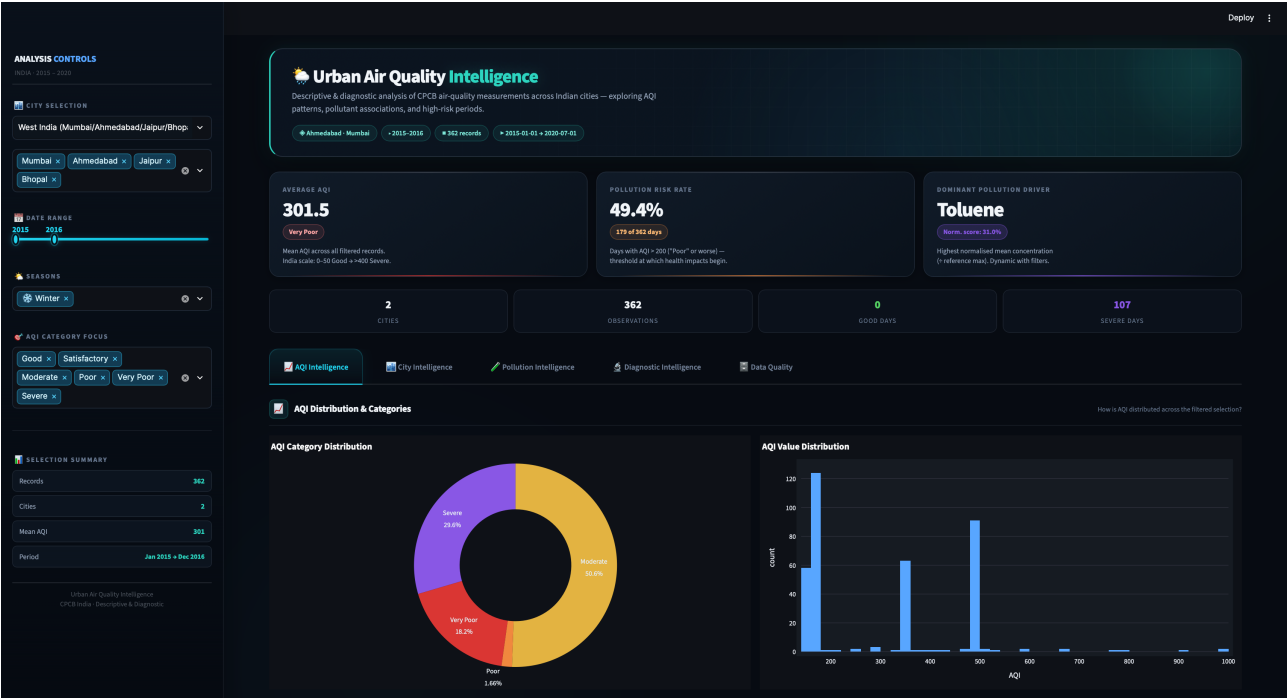


Figure 7: Dynamic filter - West India preset, 2015–2016, Winter. Dominant driver shifts from PM10 to Toluene, reflecting industrial solvent emissions in western cities

KEY OBSERVATIONS FROM THE FILTER EXAMPLES

The two filter examples above reveal analytically significant contrasts that the sidebar makes immediately discoverable:

Observation	East India 2017 Winter	West India 2015–16 Winter
Average AQI	295.7 - Poor	301.5 - Very Poor
Pollution Risk Rate	81.6% (4 in 5 days hazardous)	49.4% (1 in 2 days hazardous)
Good Days	0	0
Dominant Driver	PM10 - vehicular & industrial dust	Toluene - industrial solvents & petrochemicals

The shift in the dominant driver from PM10 (East India) to Toluene (West India) is particularly notable - it reflects a genuine difference in the emission profiles of these regions rather than a data artefact. This kind of region-specific insight is made instantly accessible by the dashboard's dynamic normalised driver computation.

- **AQI Intelligence Tab**

The AQI Intelligence tab is the primary analytical module of the dashboard. It answers the foundational question: how is air quality distributed, and how has it changed over time? The tab is structured into **two major sections: AQI Distribution & Categories and AQI Trends Over Time**, each containing multiple Plotly charts and data tables that respond dynamically to the sidebar filters.

SECTION 1 - AQI DISTRIBUTION & CATEGORIES

The first section answers: 'How is AQI distributed across the filtered selection?' It presents four components side-by-side - a category donut chart, a value histogram, a category breakdown table, and a worst pollution days table - all computed from the same filtered data slice.

DONUT CHART & HISTOGRAM

A Plotly donut chart shows the proportion of records across AQI categories, ordered from Good to Severe and colour-coded using the India AQI palette. Percentages are displayed directly on each segment.

A 60-bin histogram displays the distribution of AQI values, revealing the overall shape and concentration of pollution levels. In the full dataset, notable clusters appear around 150–200 and 300–400.

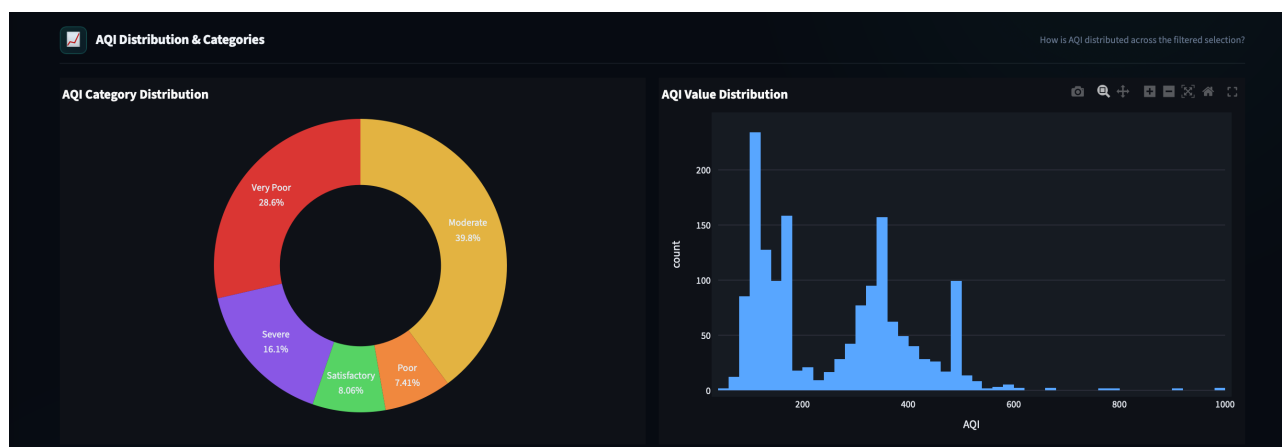


Figure 8: AQI Distribution & Categories - Left: category donut (Moderate 39.8%, Very Poor 28.6%, Severe 16.1%); Right: AQI value histogram revealing bimodal distribution with peaks at ~150 and ~350

CATEGORY BREAKDOWN TABLE

A compact AQI category table shows day counts and percentage shares, while the adjacent Worst Pollution Days table lists the 10 highest-AQI records with key pollutant values for quick comparison.

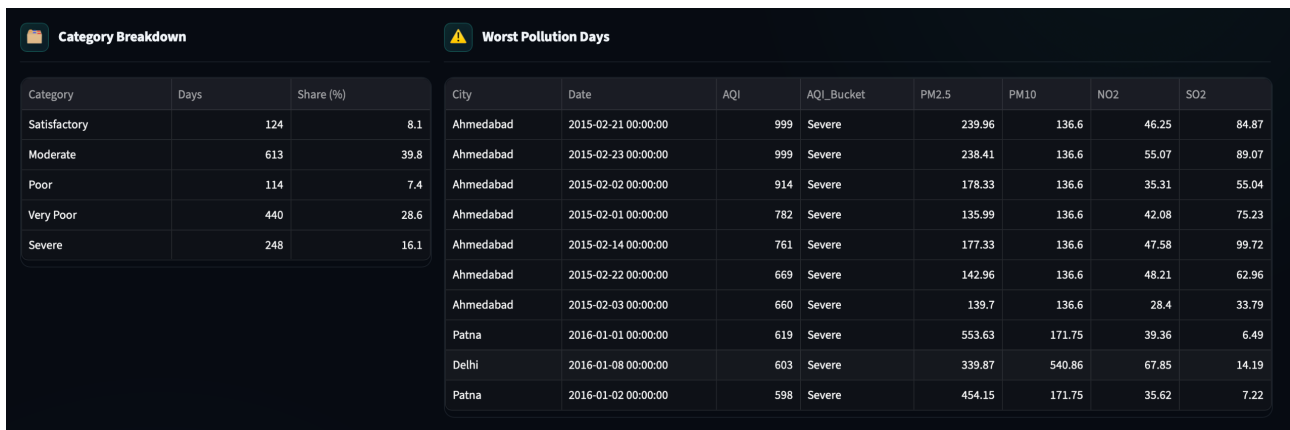


Figure 9: Category Breakdown table (left) and Worst Pollution Days table (right) - Ahmedabad dominates the worst days list in Feb 2015, followed by Patna and Delhi in Jan 2016

SECTION 2 - AQI TRENDS OVER TIME

The section examines AQI variation across years, months, and seasons using interconnected time-series and seasonal analyses.

Yearly trends show a decline in average AQI, while monthly patterns highlight higher pollution during winter months and lower levels during the monsoon. Seasonal analysis further shows a strong contrast between Winter and Summer AQI, supported by a summary table of mean AQI and record counts.



Figure 10: AQI Trends Over Time - yearly decline (2015→2020), monthly inversion-driven winter peaks, seasonal bar chart (Winter = 221.3 highest), and Seasonal Summary table

• City Intelligence Tab

The City Intelligence tab answers the comparative question: how does air quality differ across Indian cities, and how has each city's pollution evolved over time?

While the AQI Intelligence tab shows the aggregate picture for the selected filter, this tab disaggregates it by geography - ranking cities by their pollution burden, quantifying variability within each city, and tracking individual city trajectories year by year through a fully annotated heatmap.

The tab contains **two primary visualisations - the City AQI Ranking bar chart and the City × Year AQI Heatmap - plus an expandable full city summary statistics table**. Both charts use the RdYlGn_r (red–yellow–green reversed) colour scale to provide instant visual encoding of pollution severity.

CITY AQI RANKING - BAR CHART WITH ERROR BARS

The `chart_city_aqi_bar()` function renders a Plotly Express bar chart showing all cities sorted by descending mean AQI. Two pieces of information are encoded per city:

Visual Encoding	What It Communicates
Bar height + RdYlGn_r fill colour	Mean AQI - red bars signal the most polluted cities (Ahmedabad, Delhi), green bars signal the cleanest (Aizawl, Shillong, Coimbatore)
Black error bars (± 1 SD)	Variability of AQI within each city. A tall error bar means the city has extreme pollution events alongside cleaner periods; a short bar means consistently uniform air quality



Figure 11: City AQI Ranking - all 26 cities sorted by mean AQI. Ahmedabad (red, ~420) towers over Delhi (~260) and the rest. Aizawl and Shillong are the cleanest at ~20–35. Error bars show high within-city variability for Ahmedabad and Delhi

The ranking chart exposes a sharp and statistically significant geographic divide. All top-5 most polluted cities are in the North-Central Indo-Gangetic Plain (Ahmedabad, Delhi, Gurugram, Patna, Lucknow). All top-3 cleanest cities are in hilly or coastal South/Northeast India (Aizawl, Shillong, Coimbatore). This north-south gradient is driven by several compounding factors:

Factor	Effect on Northern Cities
Geography / topography	The Indo-Gangetic Plain is flat and enclosed by the Himalayas to the north, trapping pollutants in winter inversions with minimal dispersion
Industrial density	Heavy industry (steel, power plants, chemicals) concentrated in North and East India - Talcher, Jorapokhar, Brajrajnagar are coal-belt cities
Crop residue burning	Stubble burning in Punjab/Haryana (Oct–Nov) is a major seasonal PM2.5 contributor affecting Delhi, Gurugram, Amritsar
Vehicle & population density	Delhi NCR has the highest vehicle population in India; Patna and Lucknow show rapid urbanisation without corresponding emission controls

CITY × YEAR AQI HEATMAP

The **chart_city_heatmap()** function constructs a pivot table (City × Year → mean AQI) using pandas groupby, then renders it as a Plotly annotated heatmap. Each cell is labelled with the mean AQI value rounded to the nearest integer. The RdYlGn_r colour scale provides immediate visual scanning: **deep red cells are the worst (high AQI), and deep green cells are the cleanest. Black cells indicate that monitoring data was not yet available for that city-year combination.**

Dimension	Reading
Row (Y-axis)	Each row is a city - all 26 cities are listed alphabetically on the Y-axis
Column (X-axis)	Each column is a year from 2015 to 2020
Cell colour + number	Mean AQI for that city in that year. Deep red = Severe/Very Poor, yellow = Moderate/Poor, deep green = Good/Satisfactory. Black (0) = no data available for that year



Figure 12: City x Year AQI Heatmap - Ahmedabad's 2018 cell (571, deep red) is the dataset's most extreme annual average. Delhi remains yellow-orange every year. South Indian cities are consistently deep green. Black cells mark years without monitoring data

FULL CITY AQI SUMMARY STATISTICS TABLE

Beneath the heatmap, an expandable section labelled 'Full City AQI Summary Table' (implemented with `st.expander`) reveals a **six-metric per-city statistics table** produced by the `city_aqi_summary()` function. Columns include **Mean, Median, Std (standard deviation), Min, Max, and Count** - providing a complete descriptive statistical profile of each city's AQI distribution.

Full City AQI Summary Table						
City	Mean	Median	Std	Min	Max	Count
Ahmedabad	418.5	403	213.8	48	999	2009
Delhi	258.8	257	119.7	29	716	2009
Gurugram	224.6	208	116.3	38	891	1679
Patna	221.6	184	119.4	53	619	1858
Lucknow	219.5	203	115.5	39	707	2009
Talcher	171.7	131	101.1	13	570	925
Jorapokhar	151	124	75.5	27	604	1169
Brajrajnagar	148.2	120.5	65.1	22	355	938
Guwahati	140	97	112.5	25	956	502
Kolkata	135.1	86	102.4	26	475	814

Figure 13: Full City AQI Summary Statistics Table

• Pollution Intelligence Tab

The Pollution Intelligence tab shifts focus from AQI as an aggregate index to the individual pollutants that drive it. It answers three layered questions: Which pollutants contribute most to overall pollution load? How strongly does each pollutant correlate with the AQI score? And how does each pollutant's concentration vary across cities?

The tab is built around three components: **a normalised radar chart, a Pearson correlation bar chart, and a pollutant-switchable city-level box plot - plus an expandable full descriptive statistics table.**

SECTION 1 - POLLUTANT PROFILE

The Pollutant Profile section combines a **normalised contribution radar** with a **pollutant-AQI correlation bar chart**, both dynamically updated with the selected filters. The radar compares all 12 pollutants on a common 0–1 scale using reference ceilings, highlighting dominant contributors such as PM10, NO₂, NO_x and PM2.5. The correlation chart uses Pearson's r to show each pollutant's statistical association with AQI. Together, the charts reveal both the relative pollutant load and which pollutants are most closely associated with AQI variation.

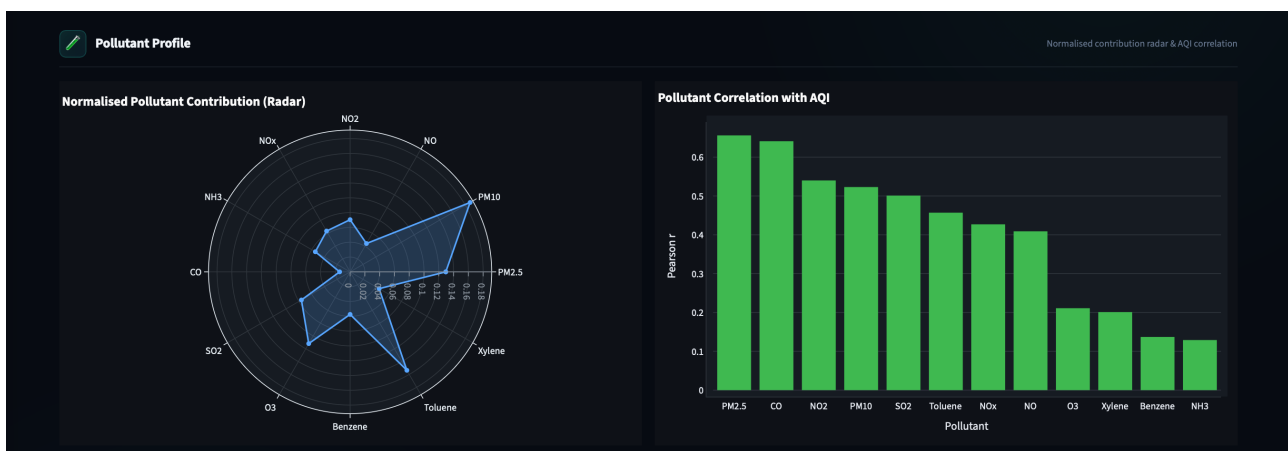


Figure 14: Pollutant Profile - Left: Normalised Contribution Radar showing PM10 with the largest spike (~ 0.18) followed by NO₂ and NO_x; Right: Pollutant Correlation bar chart with PM2.5 ($r \approx 0.67$) and CO ($r \approx 0.65$) as the strongest AQI predictors

SECTION 2 - POLLUTANT DISTRIBUTION BY CITY

The section provides a city-level distribution analysis for any selected pollutant across the filtered cities. A **dropdown allows users to choose from all 12 pollutants**, with PM2.5 selected by default. The corresponding box plot updates dynamically, ranking cities by descending median concentration to highlight the highest-concentration cities first.

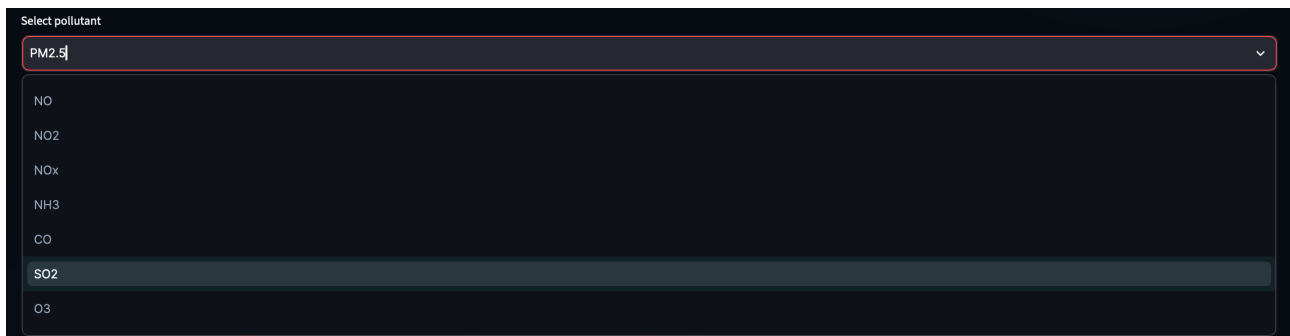


Figure 15: Pollutant selector dropdown - all 12 pollutants available.

POLLUTANT DISTRIBUTION EXAMPLES - BOX PLOT

Example A - PM2.5 Distribution:

The PM2.5 box plot displays the full daily concentration distribution for each city, ordered by descending median PM2.5. Each box shows the minimum, Q1, median, Q3, and maximum, with individual points highlighting outliers. Delhi, Gurugram, Lucknow, and Patna show higher typical concentrations, while several southern and northeastern cities exhibit lower and tighter distributions.

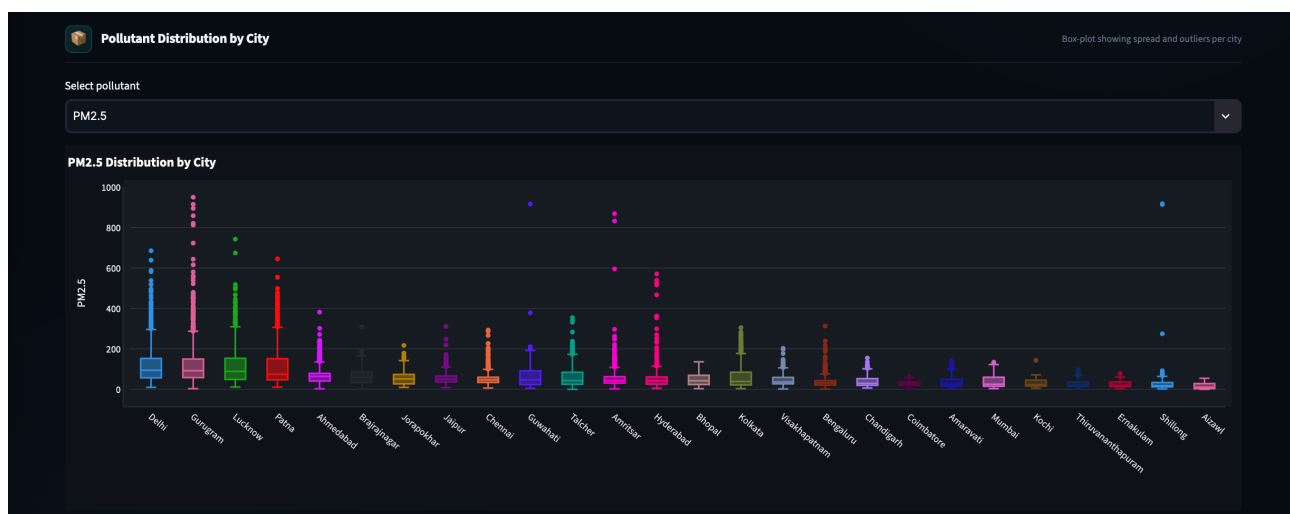


Figure 16: PM2.5 Distribution by City - Delhi, Gurugram, Lucknow, Patna lead with high median boxes (100–200 $\mu\text{g}/\text{m}^3$). Outlier dots reaching 600–1000 $\mu\text{g}/\text{m}^3$ visible for Delhi, Gurugram, Patna. Southern cities compressed near zero

Example B - SO₂ Distribution:

Switching the selector to SO₂ produces a different city ranking, highlighting pollutant-specific patterns. Ahmedabad, Talcher, and Jorapokhar show higher SO₂ concentrations, while Delhi ranks much lower than it does for PM2.5. This interactive comparison reveals distinct pollution profiles across cities from a single dashboard view.

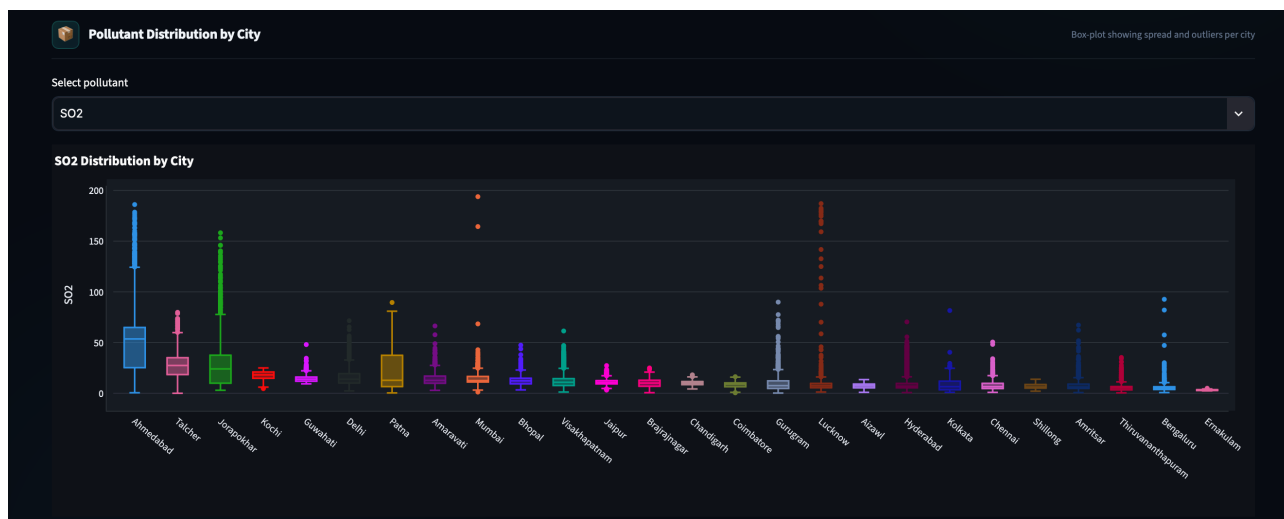


Figure 17: SO2 Distribution by City - Ahmedabad leads (median ~50, outliers ~190 $\mu\text{g}/\text{m}^3$), followed by Talcher and Jorapokhar (coal-belt cities). Delhi drops to rank 6, revealing that its pollution is PM-driven, not SO2-driven

POLLUTANT DESCRIPTIVE STATISTICS TABLE

An expandable `st.expander` labelled 'Pollutant Descriptive Statistics' reveals a transposed `pandas .describe()` table with five columns per pollutant: **Mean**, **Std Dev**, **Min**, **Median (50th percentile)**, and **Max**. All values are rounded to two decimal places. This table is the most compact way to compare the absolute magnitude, spread, and extreme values of all 12 pollutants side by side in a single view.

Pollutant Descriptive Statistics					
	Mean	Std Dev	Min	Median	Max
PM2.5	64.72	60.96	0.04	47.74	949.99
PM10	112.48	78.15	0.01	97	1000
NO	17.69	21.67	0.02	10.58	390.68
NO2	28.2	23.8	0.01	21.58	362.21
NOx	31.85	29.8	0	24.36	467.63
NH3	21.74	22.17	0.01	16.27	352.89
CO	2.55	7.04	0.01	0.97	175.81
SO2	15.21	18.24	0.01	9.74	193.86
O3	33.59	20.83	0.01	30.05	257.73
Benzene	2.87	6.48	0.01	1.37	100

Figure 18: Pollutant Descriptive Statistics Table

• Diagnostic Intelligence Tab

The Diagnostic Intelligence tab forms the statistical core of the dashboard, examining relationships between pollutants and AQI. It combines **correlation analysis, risk-day comparisons, pollutant drilldowns, anomaly detection, and driver-shift analysis**. All results are dynamically recalculated according to the active dashboard filters. The tab helps identify statistical associations, elevated pollution patterns, and changes in dominant pollutant behaviour.

INSIGHTS SUMMARY

Auto-generated insight cards highlight the most important statistical findings and patterns from the current selection.

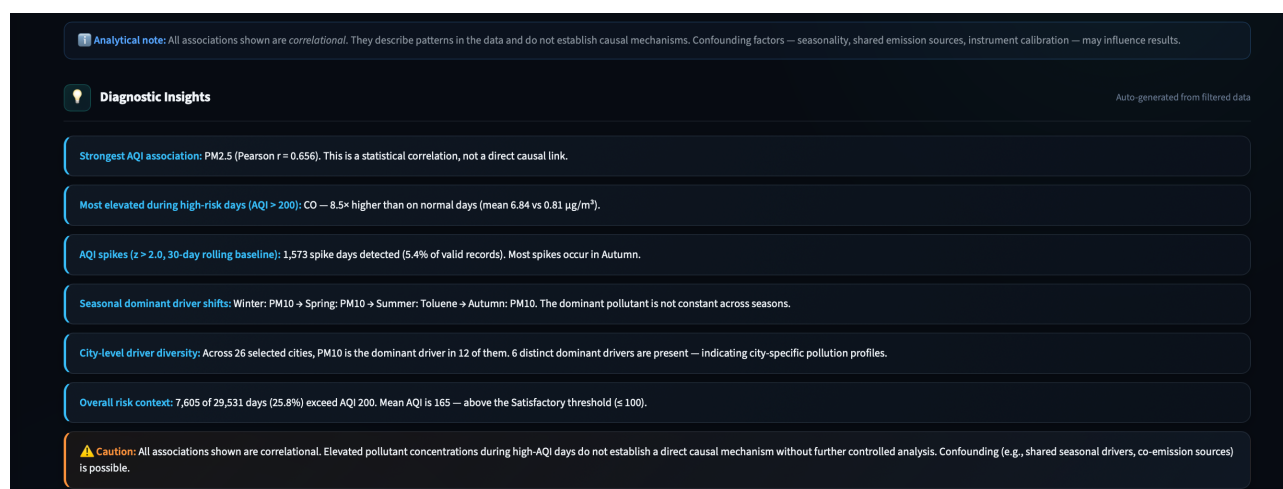


Figure 19: Auto-generated plain-language insight cards summarising key statistical findings for the current filter

POLLUTANT-AQI CORRELATION & CORRELATION MATRIX



Figure 20: Per-pollutant Pearson r bar chart (with p-values) and the full 13×13 correlation matrix heatmap

Pearson correlation results and the 13×13 heatmap show relationships among the 12 pollutants and AQI.

NORMAL VS RISK-DAY COMPARISON

Compares mean pollutant concentrations between normal days (AQI ≤ 100) and high-risk days (AQI > 200), supported by a Mann-Whitney U test.

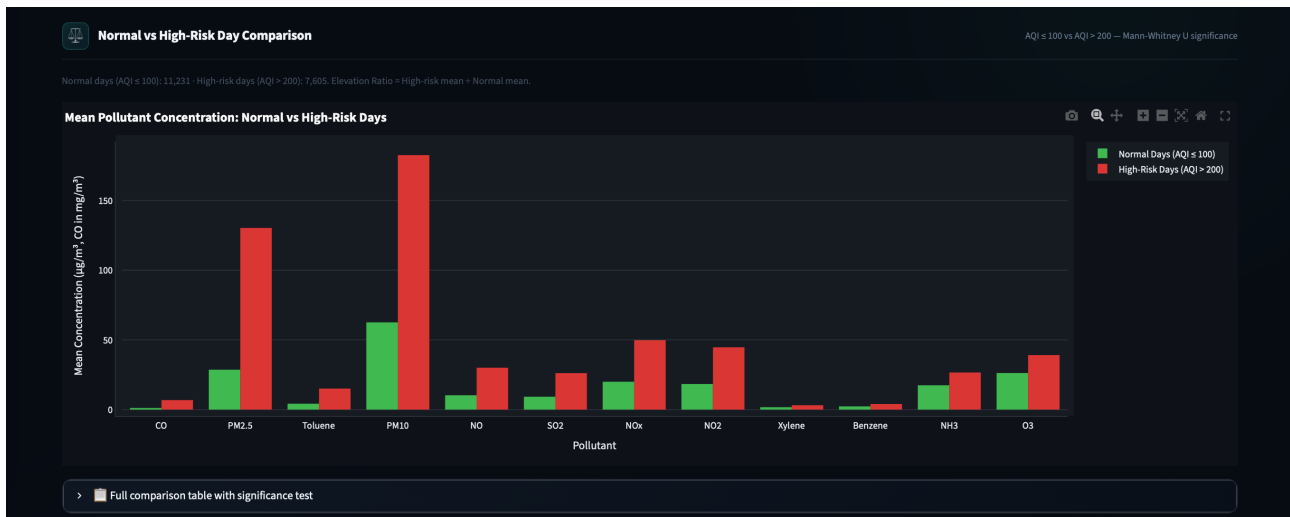


Figure 21: Mann–Whitney U test comparing pollutant concentrations on normal days (AQI ≤ 100) vs risk days (AQI > 200)

POLLUTANT-TO-AQI DRILLDOWN

Shows how pollutant concentrations vary across AQI categories, with median and mean values for the selected pollutant.

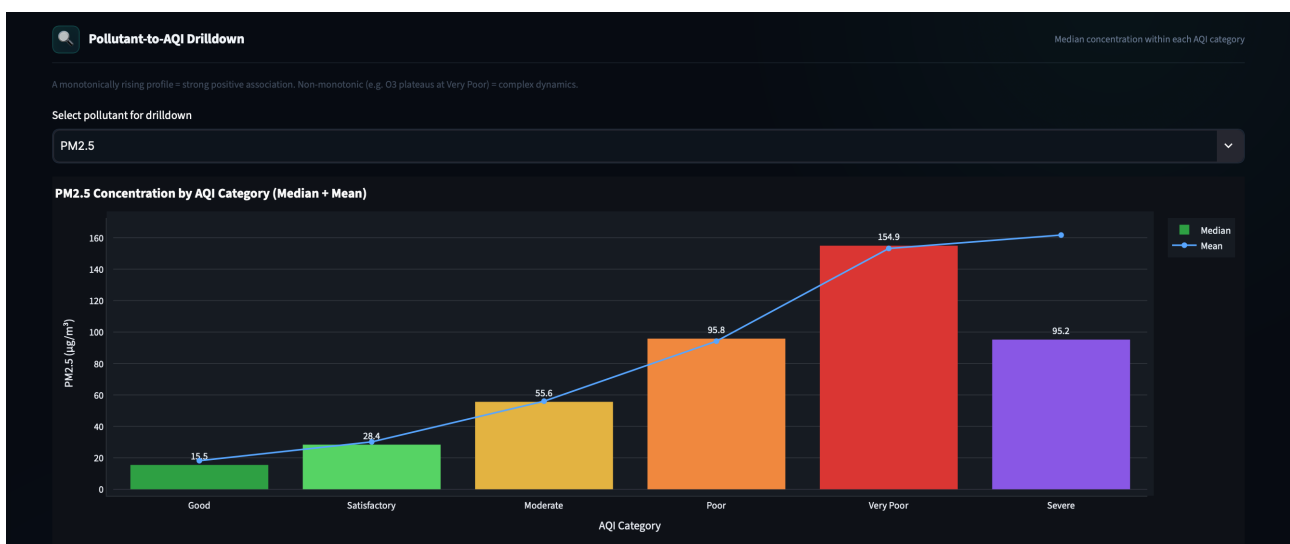


Figure 22: Scatter drilldown - individual pollutant concentration vs AQI with trend line for the selected filter

AQI SPIKE & ANOMALY ANALYSIS

Identifies unusual AQI events using a 30-day rolling z-score threshold of >2.0 and examines pollutant elevation during spike days.

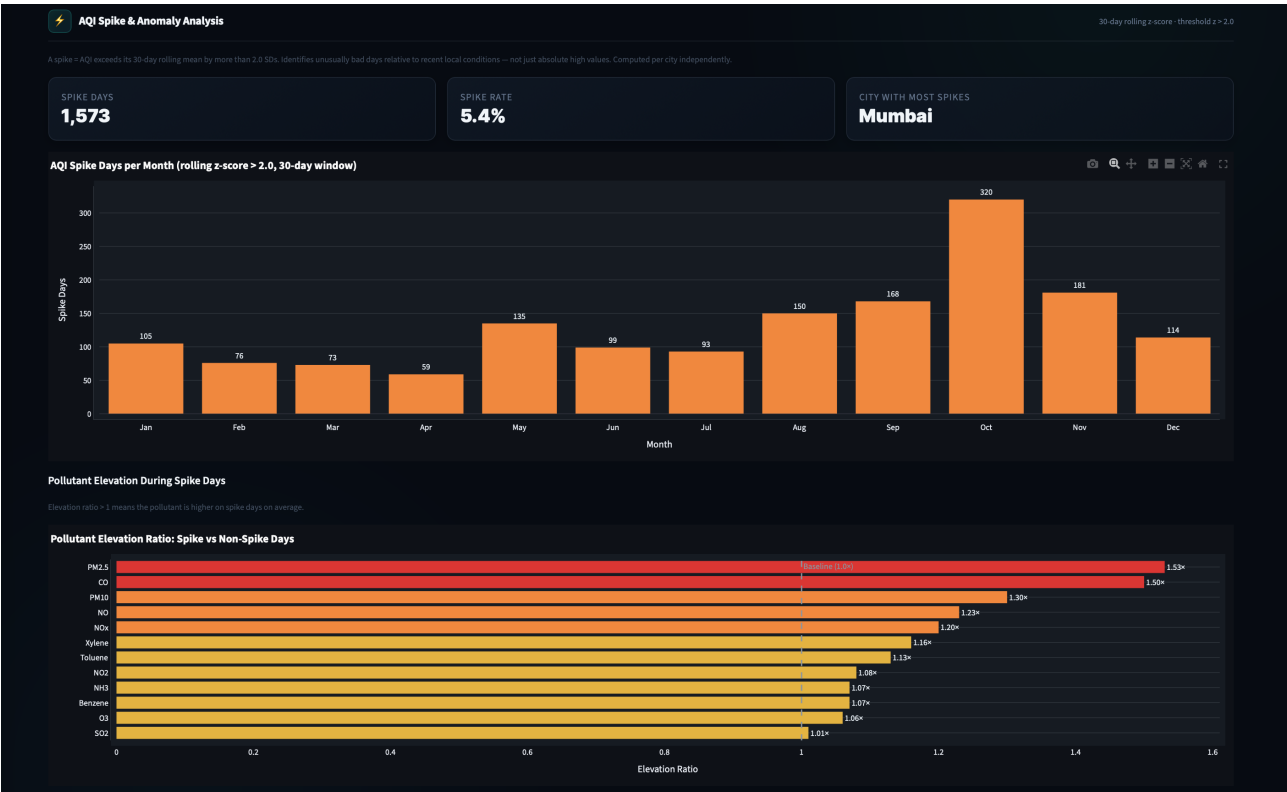


Figure 23: 30-day rolling z-score spike detection (threshold > 2.0) highlighting pollution anomaly events

DOMINANT DRIVER SHIFTS

Shows how the pollutant with the highest normalised concentration changes across seasons, years, and cities.



Figure 24: Stacked area chart showing how the dominant pollution driver shifts across years and seasons

- ## Data Quality Tab

Full data-quality transparency is exposed in a dedicated tab, covering the cleaning pipeline steps and the original missing value rates per column and city.

CLEANING PIPELINE

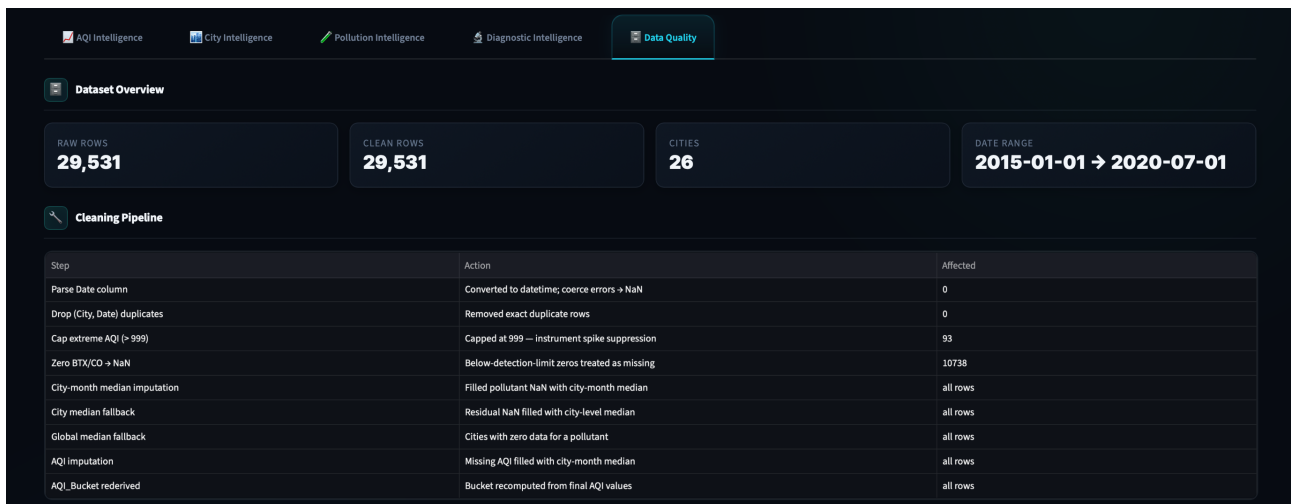


Figure 25: Step-by-step cleaning pipeline: validation → imputation → feature engineering, with record counts at each stage

ORIGINAL MISSING VALUE RATES

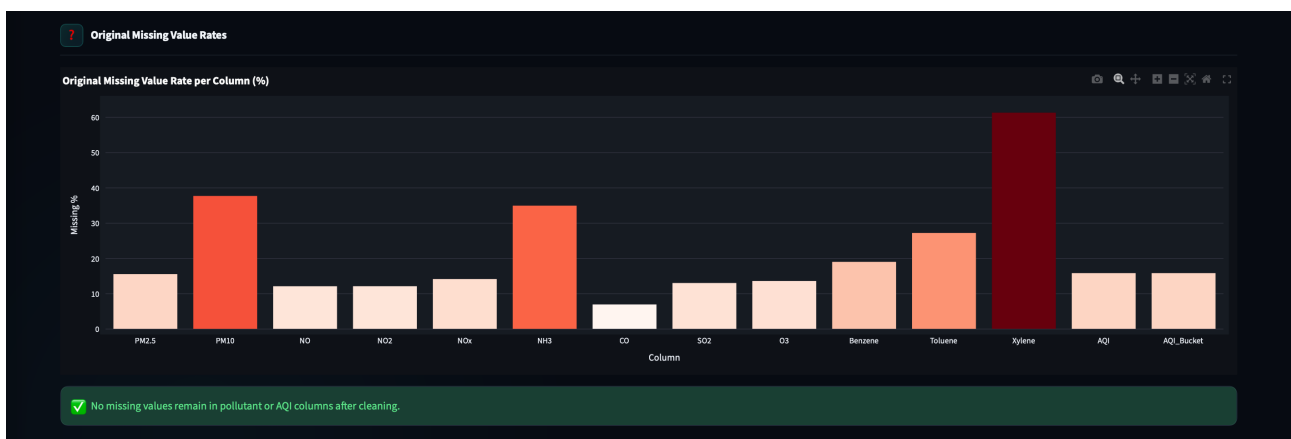


Figure 26: Per-column and per-city original missing value rates before any imputation.

05 · Conclusion

The Urban Air Quality Intelligence dashboard transforms the CPCB city-day dataset into an interactive analytical platform for exploring AQI patterns, pollutant relationships, city-level differences, seasonal behaviour, and pollution anomalies. Through descriptive visualisation and statistical diagnostics, the project provides a clearer understanding of how air quality varies across Indian cities and time periods. The interactive filtering and drill-downs make the analysis easier to explore and interpret.